Quiz 3 (Version 3)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this quiz.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not receive credit if you don't show your work.

1 Linear Transformations

Determine whether the following linear transformation is one-to-one but not onto, onto but not one-to-one, both, or neither. Show your work and justify your answer.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + 3x_2 \\ -x_1 - x_2 + 4x_3 \\ -2x_2 - 4x_3 \\ x_1 + 7x_2 + 8x_3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 0 \\ -1 & -1 & 4 \\ 0 & -2 & -4 \\ 1 & 7 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 \\ 0 & 2 & 4 \\ 0 & -2 & -4 \\ 0 & 4 & 8 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Solution.

◯ one-to-one, not onto ◯ onto, not one-to-one ◯ both ● neither

Justification.

2 pivots, 4 rows
3 columns

2 Matrix Inverses

Determine the inverse of the following matrix and use it to solve the equation $A\mathbf{x} = \mathbf{b}$. You may not solve $A\mathbf{x} = \mathbf{b}$ by row-reducing the augmented matrix $[A\ \mathbf{b}]$.

$$A = \begin{bmatrix} 1 & -3 & -1 \\ 1 & -2 & 1 \\ 1 & -3 & 0 \end{bmatrix} \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 1 & -3 & -1 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 1 & 0 \\ 1 & -3 & 0 & 0 & 0 & 1 \end{array}\right] \sim \left[\begin{array}{ccc|ccc} 1 & -3 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -3 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 3 & -5 \\ 0 & 1 & 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\vec{x} = A^{-1}\vec{b} = \begin{bmatrix} 3 & 3 & -5 \\ 1 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Solution.

$$A^{-1} = \begin{bmatrix} 3 & 3 & -5 \\ 1 & 1 & -2 \\ -1 & 0 & 1 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

3 Matrix Equations

Determine the matrix X such that $AXB^T = I$.

$$A = \begin{bmatrix} -1 & 1 \\ -1 & 2 \end{bmatrix} \qquad B = \begin{bmatrix} -1 & -1 \\ -2 & -1 \end{bmatrix}$$

$A^{-1} = \frac{1}{-2-(-1)} \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ -1 & 1 \end{bmatrix}$

$B^T = \begin{bmatrix} -1 & -2 \\ -1 & -1 \end{bmatrix} \quad (B^T)^{-1} = \frac{1}{1-2} \begin{bmatrix} -1 & 2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$

$X = A^{-1}(B^T)^{-1} = \begin{bmatrix} -2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ -1 & 1 \end{bmatrix}$

Solution.

$$\begin{bmatrix} -3 & 5 \\ -2 & 3 \end{bmatrix}$$

4 LU Factorization

Consider the following partial implementation of the function `lu_2_by_2` that, given a (2×2) NumPy array, returns an LU factorization of it.

```
def lu_2_by_2(a):
    assert(a.shape == (2, 2) and not np.isclose(a[0, 0], 0.) and a.dtype == 'float64')

    u = np.copy(a)
    l = np.eye(2)

    c = ???
    u[1] += c * u[0]
    l[1, 0] = -c

    return (l, u)
```

Determine an expression for `???` that completes the implementation.

Solution.

$$-a[1,0] \,/\, a[0,0]$$